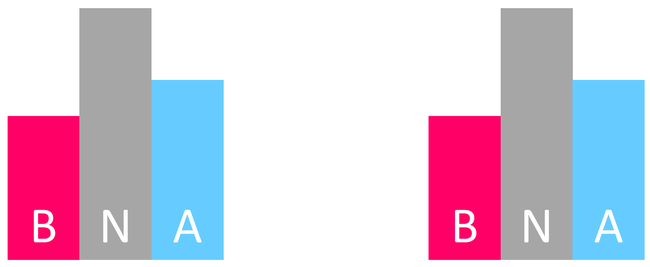
利用長期天氣展望與其不確定性之流量超越機率推估
—以石門水庫入流量為例

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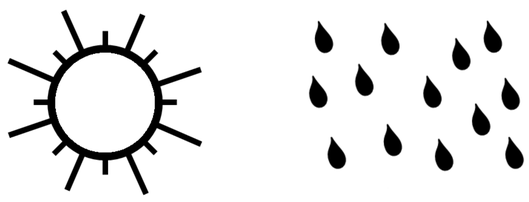
中央氣象局(CWB)的長期天氣展望
高於正常(A)、正常(N)、低於正常(B)
三類別可能性



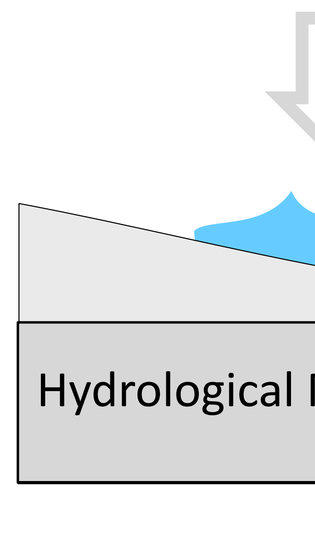
Pr(class|outlook)

Monte Carlo simulation

Weather Generator

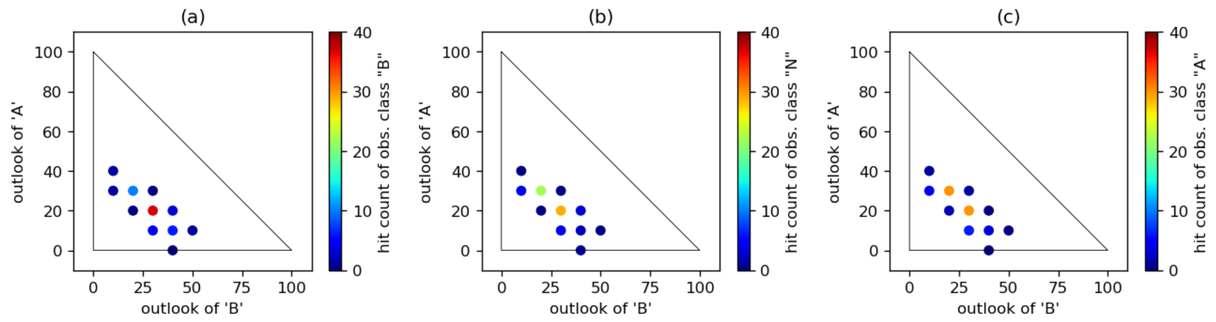


Number of the weather series belonging to each category is generated in the same ratio to the posterior probabilities (Pr(class|outlook)).



流量超越機率

展望與實際類別命中，隱含**預報不確定性**



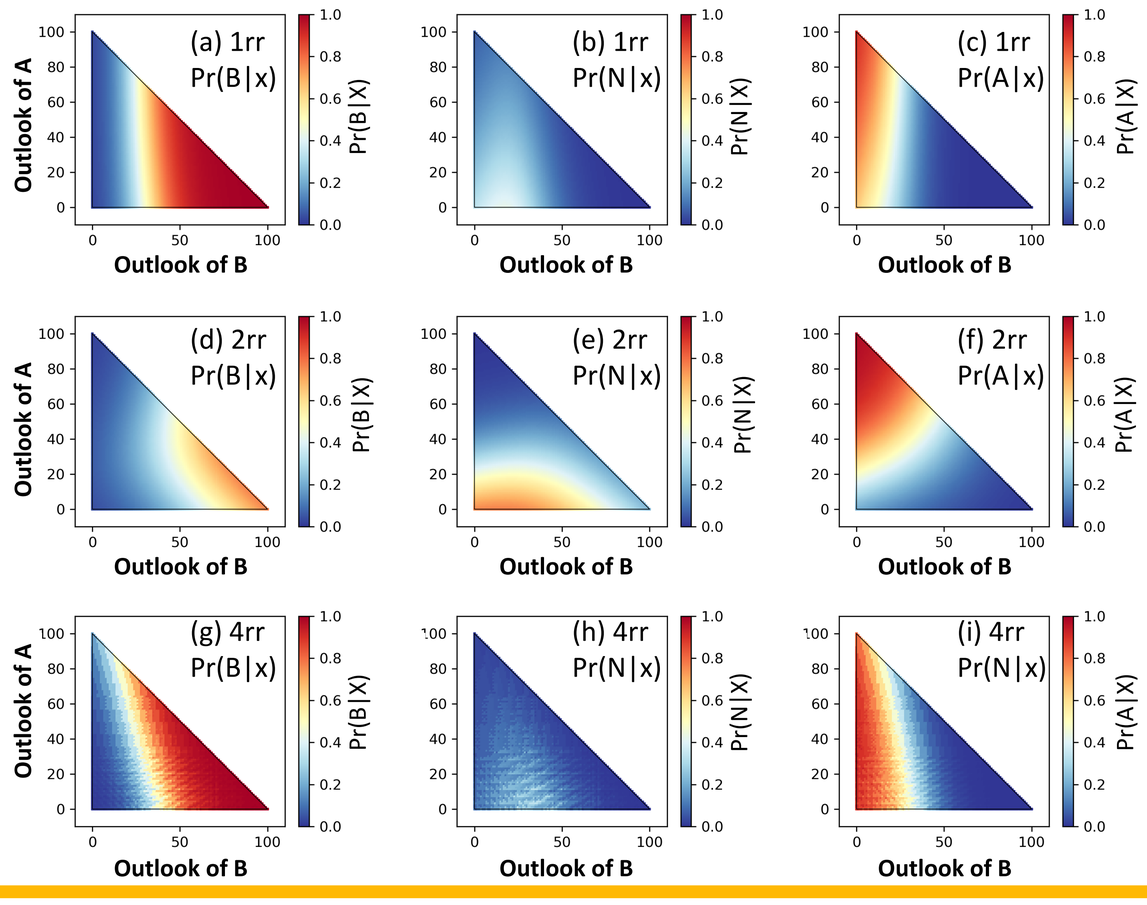
Logistic Regression

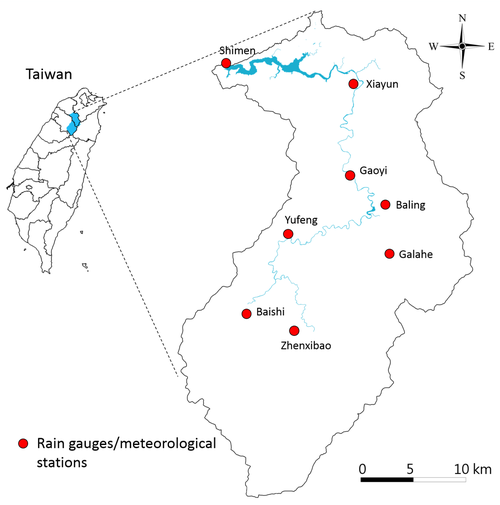
$$\ln\left[\frac{\Pr(G=k|X)}{\Pr(G=K|X)}\right]=\beta_{0k}+\beta_{1k}^{T}X,\quad k=1,\ldots,K-1$$

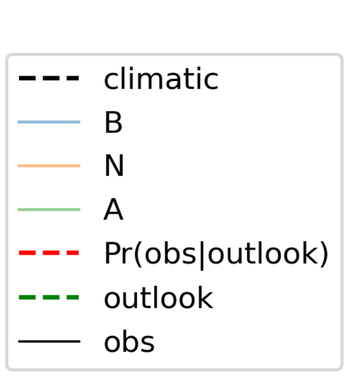
$$\Pr(G=k|X=x)=\frac{\exp\left(\beta_{k0}+\beta_{k}^{T}x\right)}{1+\sum_{l=1}^{K-1}\exp\left(\beta_{l0}+\beta_{l}^{T}x\right)},\quad k=1,..,K-1$$

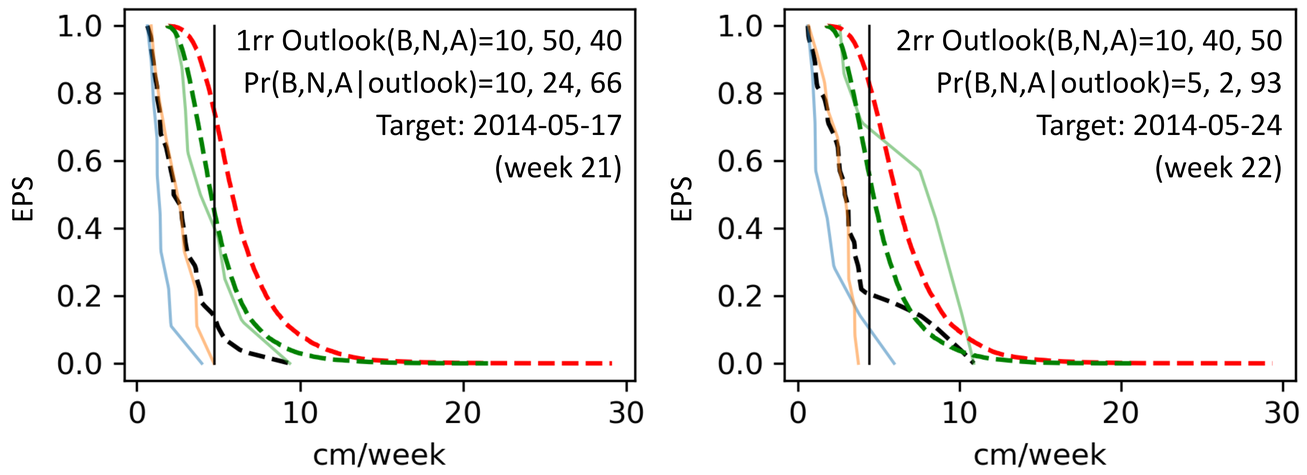
$$\Pr(G=K|X)=\frac{1}{1+\sum_{k=1}^{K-1}\exp\left[\beta_{0k}+\beta_{1k}^{T}X\right]}$$

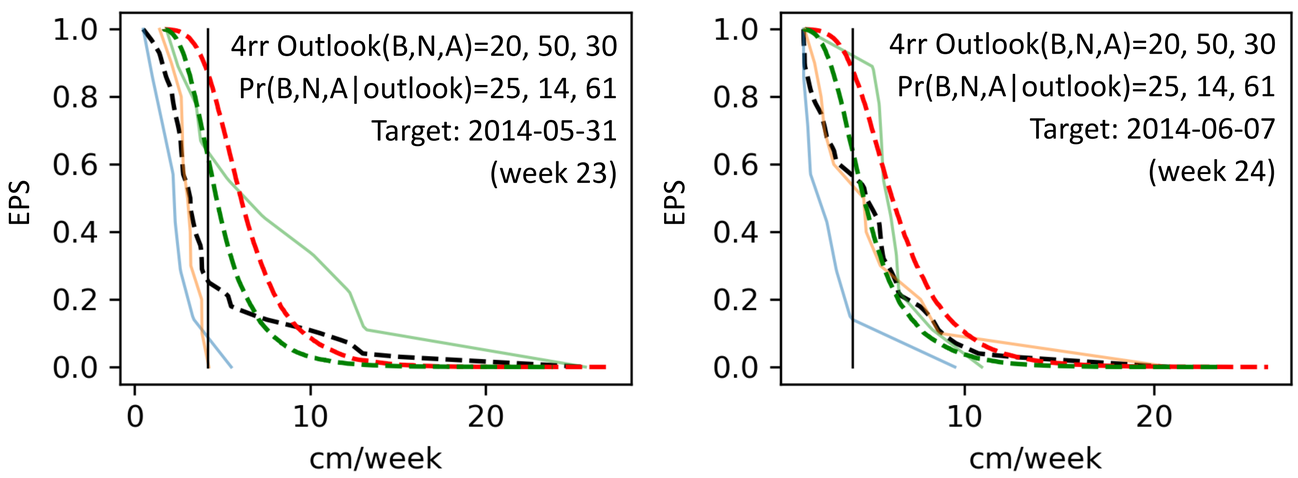
β_{0k} and β_{1k}^{T} are regression coefficients. Number of classes is 3 (K=3).
Predictor X is the likelihood indicated by the outlook.











石門水庫為研究地區，我們訓練了Logistic Regression 模型，據此計算入流量超越機率(EPS)。以 2014年5月16日(周五), 發布之雨量月展望(未來第一周、第二周、第三四周類別可能性)為例。此計算結果認為未來降雨偏高可能性高，並得高流量可能性較傳統方法(長期觀測建立機率取縣)、與CWB預報加權平均曲線高。此方法合理地納入了預報不確定性，但並不代表具有更精準預報能力，其預報能力仍受限天氣展望此產品本身預報是否能抓住未來趨勢。